Reflexivity of sequent derivability

The Open Logic Project contributors

openlogic:first-order-logic/sequent-calculus:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author proposition labelled prop:reflexivity (706-709); the proof span 711-714 is the author proof environment that immediately follows it.

Editorial note. No cross-references occur in the retained span.

Editorial scope. One original result of the Open Logic Project unit first-order-logic/sequent-calculus, counted once: reflexivity of derivability. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Initial sequents. The sequent !A sequent !A is an initial sequent and hence derivable in the sequent calculus.. It supplies the derivation used in the proof (712-713).
- Derivability from a set of sentences. Gamma proves !A when some finite Gamma_0 subset of Gamma has the sequent Gamma_0 sequent !A derivable.. The proof closes the gap between the initial sequent and derivability from the arbitrary set Gamma (712-714).

Proposition 0.1 (Reflexivity)**.** *If* $!A \in \Gamma$*, then* $\Gamma \vdash !A$*.*

Proof. The initial sequent $!A \Rightarrow !A$ is !!derivable, and $\{!A\} \subseteq \Gamma$. □

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